

Instruction Set Architecture of Qiuniu

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Abstract

Qiuniu is the first-generation atom array quantum computation platform at AtoMatrix of Tsinghua and at Qosmos, which is now fully functioning. This note summarizes the operations of different levels (a.k.a. instruction set architecture, ISA) that can be carried out on this platform, with an estimated cost (time and fidelity) given. This instruction set serves as a reference for theorists who are interested in developing quantum algorithms, either in analog or in digital form, based on atom arrays.

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1. Hardware pulse-level operations

1.1. Static tweezer configuration

One can define a set of static tweezers: $\vec{r}_i^{(s)} = (x_i^{(s)}, y_i^{(s)})$; $i = 1, 2, 3, \dots, N^{(s)}$, where $N^{(s)}$ is the number of static tweezers. The tweezers can be placed as close as $\sim 2.5 \mu\text{m}$ to each other, with an accuracy of $\sim 0.1 \mu\text{m}$.

1.2. System model

The Hamiltonian that is natural to the atom array platform is

$$\hat{H} = \frac{\Omega(t)}{2} \sum_i \hat{\sigma}_i^{(x)} - \sum_i (\Delta_b(t) + \alpha_i \Delta_s(t)) \hat{n}_i + \sum_{i < j} V_{ij} \hat{n}_i \hat{n}_j. \quad (1)$$

Here, $\hat{\sigma}^{(x)} = |0\rangle\langle 1| + |1\rangle\langle 0|$, $\hat{n} = |1\rangle\langle 1|$. $V_{ij} = C_6 \cdot \left| \vec{r}_i^{(s)} - \vec{r}_j^{(s)} \right|^{-6}$ is the Van der Waals interaction. C_6 can be chosen in the range $30 \sim 500 \text{ GHz} \cdot \mu\text{m}^6$.

$\Omega(t)$ is the global driving coupling strength, which can be dynamically tuned with a time resolution of $\sim 1 \text{ ns}$, a rising/falling time of $\sim 20 \text{ ns}$, and a maximum value of $\sim 10 \text{ MHz}$.

$\Delta_b(t)$ is the global detuning bias, which can be dynamically tuned with a time resolution of $\sim 1 \text{ ns}$, a rising/falling time of $\sim 20 \text{ ns}$, and a maximum value of $\sim 20 \text{ MHz}$.

$\alpha_i \Delta_s(t)$ is the site-wise detuning, which can be dynamically tuned with a time resolution of $\sim 1 \text{ ns}$, a rising/falling time of $\sim 20 \text{ ns}$, and a maximum value of $\sim 10 \text{ MHz}$. α_i can be set with a relative accuracy of $\sim 5\%$.

The whole time series can be as long as $\sim 10 \mu\text{s}$.

2. A self-contained description of near-term accessible operations

2.1. Atom movement

See animation at this [website](#). The notation there is not exactly the same as that here, but the idea is the same.

2.1.1. Lattice configuration

The legitimate positions (sites) of atoms are defined by a rectangular $N_x^{(s)} \times N_y^{(s)}$ lattice

$$\left\{ x_1, x_2, \dots, x_{N_x^{(s)}} \right\}, \left\{ y_1, y_2, \dots, y_{N_y^{(s)}} \right\}. \quad (2)$$

The entire lattice can be divided into several zones, e.g., 1Q zone, 2Q zone, and Meas zone. For $\forall y, y \in y[\text{some zone}]$. Each function can have several zones (though not favorable). For example, there can be multiple 1Q zones. Zones can overlap with each other (though not favorable). The region of zones can be set at will, but once set, it cannot be changed during a circuit.

Either a static tweezer or a mobile tweezer must correspond to a lattice site. Therefore, each lattice site can hold at most two atoms, one trapped by a static tweezer, and the other by a mobile tweezer. The static tweezers are on for all time, while the mobile tweezers are functioning in the form of motion cycle. The occupation of the lattice is kept as two matrices $O^{(s)}[i, j]$ and $O^{(m)}[p, q]$.

2.1.2. Mobile tweezer motion cycle

However, it is optimal to use the mobile tweezer only for translations. And it is optimal to set mobile tweezer lattice to be equally spaced. Due to hardware limitations, equal spacing, and compression or expansions, of the mobile tweezer lattice can cause significant decoherence and errors.

Therefore, the mobile tweezers now form a rectangular $N_x^{(m)} \times N_y^{(m)}$ sub-lattice $L^{(m)}$ of the site lattice, with fixed equal spacing. The definition is thus more simple,

$$L^{(m)} = (x_1, y_1, a_x, a_y), \quad (3)$$

where x_1 (y_1) are integers in $1, 2, \dots, N_x^{(s)}$ ($N_y^{(s)}$), which mark the position of the upperleft mobile tweezer; a_x (a_y) are integers, which mark the lattice constant of the mobile tweezers.

The motion cycle is then defined as a time series of the reference labels (3), which is a $(T + 1) \times 2$ matrix

$$M(N_x^{(m)}, N_y^{(m)}, T) = \begin{pmatrix} x_1(0) & y_1(0) \\ x_1(1) & y_1(1) \\ \vdots & \vdots \\ x_1(T) & y_1(T) \end{pmatrix} \quad (4)$$

corresponding to T movement steps.

The loading operation updates the occupation matrices by

$$\begin{aligned} O^{(m)}[p, q] &\Leftarrow O^{(s)}[x_1(0) + pa_x, y_1(0) + qa_y], \forall (p, q) \in L^{(m)} \\ O^{(s)}[x_1(0) + pa_x, y_1(0) + qa_y] &\Leftarrow 0, \forall (p, q) \in L^{(m)} \end{aligned} \quad (5)$$

The end state of the motion cycle must satisfy

$$O^{(s)}[x_1(T) + pa_x, y_1(T) + qa_y] O^{(m)}[p, q] = 0, \forall (p, q) \in L^{(m)}, \quad (6)$$

so that an off-loading operation can follow the motion cycle, which update the occupation by

$$\begin{aligned} O^{(s)}[x_1(0) + pa_x, y_1(0) + qa_y] &\Leftarrow O^{(s)}[x_1(0) + pa_x, y_1(0) + qa_y] + O^{(m)}[p, q], \forall (p, q) \in L^{(m)} \\ O^{(m)}[p, q] &\Leftarrow 0, \forall (p, q) \in L^{(m)} \end{aligned} \quad (7)$$

2.2. Gate operations

Between any two atom movement steps, a gate operation can be performed.

2.2.1. 1Q gates

One can perform any native gates of two forms, for all atoms with $y \in y[1Q \text{ zone}]$ simultaneously.

1) $R_\varphi(\theta)$, where φ means rotation about axis $X \cos \varphi + Y \sin \varphi$. 2) $Z(\theta)$, which is a phase gate.

2.2.2. 2Q gates

One can perform any CPHASE(θ) gates for all atoms with $y \in y[2Q \text{ zone}]$ simultaneously, as long as those sites have double occupation held by both static and mobile tweezers. Sites with only one atom is not affected by CPHASE(θ).

2.3. Measurement

At the very end of the entire circuit, a measurement operation can be performed. One can measure an atom in computational basis $|0\rangle/|1\rangle$ for all atoms with $y \in y[\text{Meas zone}]$ simultaneously.

2.4. Cost estimates

Operation	Time cost	Fidelity
Single qubit gate	$\sim 1 \mu\text{s}$	99.95%
Two-qubit gate	$\sim 300 \text{ ns}$	99.5%
Measurement	$\sim 20 \text{ ms}$	99%
On/off loading	$\sim 100 \mu\text{s}$	~ 1
Moving	0.5 m s^{-1}	~ 1

Table 1. Table for time and fidelity for different operations

3. Version history

0.2.1

Add a link to the animation of atom movement on 2026-04-21.

- Add the animation link in the section of atom movement, so that readers can have a more intuitive understanding of how the atom movement works.

0.2.0

Add simulation level description on 2026-04-19.

- Add the Hamiltonian description of the system, allowing pulse-level design for quantum simulation, quantum optimization, quantum optimal control, etc.

0.1.1

Refine after discussion with Xiaoming Sun's Team on 2026-04-19.

- Remove the single-qubit phase acquired during two-qubit gates, since this is more simple to eliminate in experiments rather than in circuit design.
- Add an estimate of the cost of operations.

0.1.0

Initial version on 2026-03-27

- Give a self-contained description of near-term available operations.